

Oracle Cloud Infrastructure — Always Free Resources & Compute Shapes

Documentação de referência oficial (consolidada) — Base de conhecimento para
o Alura Agente

Fonte: docs.oracle.com — consolidado em agosto de 2026

Contents

Always Free Resources	1
Infrastructure	1
Database	4
Networking	4
Observability and Management	5
Additional Services	5
Service Usage and Limits	5
Compute Shapes	6
Compute Shape Pricing — OCPUs and vCPUs	6
Flexible Shapes	6
Virtual Machine (VM) Standard Shapes — Full Reference	7
GPU Shapes (VM)	8
Bare Metal Shapes — Standard Series	8
Bare Metal GPU Shapes (selection)	8
C-States and Frequency Scaling	9
Previous Generation Shapes	9
Dedicated Virtual Machine Host Shapes (selection)	9

Always Free Resources

All Oracle Cloud Infrastructure accounts (whether free or paid) have a set of resources that are free of charge in the home region of the tenancy, for the life of the account. These resources display the **Always Free-eligible** label in the Console.

Using the Always Free resources, you can provision a virtual machine (VM) instance, an Oracle Autonomous AI Database, and the networking, load balancing, and storage resources needed to support the applications that you want to build. With these resources, you can run small-scale applications or perform proof-of-concept testing.

Infrastructure

Certificates

All tenancies get five certificate authorities (CAs) and 150 certificates included in the Always Free resources.

Compute

All tenancies get a set of Always Free resources in the Compute service for creating compute virtual machine (VM) instances. You must create the Always Free compute instances in your home region.

Note: If you receive an “out of host capacity” error when trying to create a Compute instance, this indicates a temporary lack of Always Free shapes in your home region. Try creating the instance in a different availability domain, or wait a while, then try again. You can also upgrade your account to Pay as You Go, which gives you access to more types of Compute resources — Oracle doesn’t charge for Always Free resources after you upgrade, and will only charge for usage above the Always Free limits.

Available Shapes

- **Micro instances (AMD processor):** All tenancies get up to two Always Free VM instances using the VM.Standard.E2.1.Micro shape, which has an AMD processor.
- **OCI Ampere A1 Compute instances (Arm processor):** All tenancies get the first 1,500 OCPU hours and 9,000 GB hours per month for free for VM instances using the VM.Standard.A1.Flex shape, which has an Arm processor. For Always Free tenancies, this is equivalent to **2 OCPUs and 12 GB of memory**.

In regions with multiple availability domains:

- You can create OCI Ampere A1 Compute instances in any availability domain, except South Korea North (Chuncheon).
- Instances using the VM.Standard.E2.1.Micro shape can only be created in one availability domain.

Number of Compute Instances Available to Your Account Depending on the size of the boot volume and the number of OCPUs allocated to each OCI Ampere A1 Compute instance, you can create one or two OCI Ampere A1 Compute instances (2 OCPUs total). The minimum boot volume size for each instance is 47 GB, regardless of shape. Your account comes with 200 GB of Always Free block volume storage, used to create boot volumes.

For example, using the default boot volume size of 47 GB, you could provision two VM.Standard.E2.1.Micro instances and one OCI Ampere A1 Compute instance with 2 OCPUs — or two OCI Ampere A1 Compute instances with 1 OCPU each and zero VM.Standard.E2.1.Micro instances.

IP Addresses for Compute Instances You do not have to assign a public IPv4 address to every compute instance in your tenancy. You can create a compute instance in a public subnet without a public IP address, or create an instance in a private subnet.

Idle Compute Instances Important — Reclamation of Idle Compute Instances

Idle Always Free compute instances may be reclaimed by Oracle. Oracle deems VM and bare metal compute instances as idle if, during a 7-day period, the following are all true:

- CPU utilization for the 95th percentile is less than 20%
- Network utilization is less than 20%
- Memory utilization is less than 20% (applies to A1 shapes only)

Details of the Always Free Compute Instances VM.Standard.E2.1.Micro shape (AMD):

- **Processor:** 1/8th of an OCPU with the ability to burst to use additional CPU resources
- **Memory:** 1 GB

- **Networking:** One VNIC with one public IP address and up to 50 Mbps network bandwidth via the internet. Traffic to private IPs, on-premise endpoints via a Dynamic Routing Gateway, or to endpoints within the same OCI region is up to 480 Mbps.
- **Image:** Oracle Linux Cloud Developer 8, Oracle Linux, Ubuntu, or CentOS. (Oracle Linux Cloud Developer has some programs not installed due to the shape's limited memory.)

VM.Standard.A1.Flex shape (Arm-based OCI Ampere A1 Compute):

Because this is a flexible shape, you can customize the number of OCPUs and amount of memory allocated when you create or resize an instance. You can use all of the Always Free OCPUs and memory for a single instance, or create up to two instances of 1 OCPU each (2 OCPUs total).

- **Processor:** 2 OCPUs total, allocated flexibly
- **Memory:** 12 GB total, allocated flexibly
- **Networking:** Bandwidth and number of VNICs scale with the number of OCPUs
- **Image:** Oracle Linux Cloud Developer (requires at least 8 GB memory), Oracle Linux, or Ubuntu

Tip: Linux images labeled “Always Free Eligible” in the Console are compatible with Always Free compute instances and incur no licensing fees. To provision a compute instance with an image that is not Always Free-eligible, you must have a paid account or a Free Trial account with available credits.

Block Volume

All tenancies receive a total of **200 GB** of Block Volume storage, and five volume backups, included in the Always Free resources. These amounts apply to both boot volumes and block volumes combined. When you provision a compute instance, it automatically receives a 50 GB boot volume. You can also create and attach additional block volumes to expand storage capacity.

Important:

- To create an Always Free block volume, it must be created in the home region of the tenancy. Volumes created outside the home region incur regular costs.
- When an Always Free resource is deleted, capacity becomes available and Block Volume automatically tries to transition an existing paid resource to Always Free.

Because the minimum boot volume size is 50 GB, launching four instances will use all your Always Free Block Volume resources. Alternatively, you can launch one instance with the default 50 GB boot volume and attach a 150 GB block volume to expand storage. You can have a maximum of five Always Free volume backups at any time (boot and block volume backups combined).

Object and Archive Storage

All tenancies get a total of 20 GB of Always Free Object Storage.

If your Free Trial has expired and your account is Always Free only:

- 20 GB combined Standard tier, Infrequent Access tier, and Archive tier data
- 50,000 Object Storage API requests per month

If you have a paid account or Free Trial credits:

- 10 GB Standard tier + 10 GB Infrequent Access tier + 10 GB Archive tier data
- 50,000 Object Storage API requests per month

Important: If you are on a Free Trial, you can store unlimited data and the first 20 GB incurs no deduction from your \$300 trial credit. If you do not upgrade before your trial ends, your account is limited to 20 GB combined; if you are using more than 20 GB when the trial ends, all objects beyond the limit will be deleted.

Vault

All master encryption keys protected by software are free. All tenancies get 20 key versions of master encryption keys protected by a hardware security module (HSM), and 150 Always Free Vault secrets, spreadable across any number of vaults (virtual private vaults are not included). Each secret supports up to 40 secret versions (20 active + 20 pending deletion). Deleted secrets take at least a day to actually be removed and free up allotment.

Resource Manager

All tenancies get Always Free resources in Resource Manager for automating infrastructure provisioning with Terraform:

Resource	Limit Name	Always Free
Configuration source providers	configuration-source-provider-count	100
Jobs (concurrent, 24h max duration)	concurrent-job-count	2
Private endpoints	private-endpoint-count	1
Private endpoint reachable IP addresses	private-endpoint-reachable-ip-count	100
Private templates	template-count	100
Stacks (250 vars/stack, 8192 bytes/var, 11 MB zip)	stack-count	100

Database

- **Oracle Autonomous AI Database:** Two Always Free databases per tenancy, for transaction processing, data warehousing, Oracle APEX application development, or JSON-based application development.
 - Processor: 1 OCPU (cannot be scaled)
 - Storage: 20 GB (cannot be scaled)
 - Workload types: Autonomous AI Transaction Processing, Autonomous AI JSON Database, Oracle APEX Application Development, or Autonomous AI Lakehouse
 - Maximum simultaneous database sessions: 20
 - Exadata infrastructure type: Serverless
- **Oracle NoSQL Database:** Up to 133 million reads/month, 133 million writes/month, and 3 tables with 25 GB storage per table.
- **Oracle MySQL HeatWave:** A standalone MySQL HeatWave DB system with a single-node HeatWave cluster in the home region — 50 GB storage for data/logs plus an extra 50 GB of backup storage.

Networking

- **Cluster Placement Groups:** 10 to 50 Always Free cluster placement groups per region, depending on the service's pricing model.
- **Load Balancing:** Tenancies created December 15, 2020 or later get one Always Free Flexible Load Balancer (10 Mbps min/max bandwidth, 16 listeners, 16 virtual hostnames, 16 backend sets, 1024 backend servers). Tenancies created earlier get a Micro shape load balancer (10 Mbps, 10 listeners, 10 virtual hostnames, 10 backend sets, 128 backend servers).
- **Network Load Balancer:** One Always Free Network Load Balancer — 50 listeners, 50 backend sets, 512 backend servers per set, 1024 backend servers total.
- **Virtual Cloud Networks (VCNs):** Free Tier tenancies can have up to 2 VCNs, with IPv4 and IPv6 support. Tenancies cannot send email through outbound TCP port 25 to the internet by default — you must request an exemption for that.

- **VCN Flow Logs:** Free Tier tenancies receive up to 10 GB/month shared across the OCI Logging service, for auditing traffic accepted/rejected by security list rules.
- **Site-to-Site VPN:** Up to 50 IPSec connections with your Free Tier account.

Observability and Management

- **Application Performance Monitoring:** 1000 tracing events and 10 Synthetic Monitor runs per hour.
- **Connector Hub:** 2 Always Free connectors (for moving data between OCI services and third-party services).
- **Console Dashboards:** 100 dashboards per tenancy.
- **Email Delivery:** 3000 free emails per month.
- **Fleet Application Management:** Free system lifecycle management for the first 25 resources per month (Compute instances and databases).
- **Monitoring:** 500 million ingestion data points and 1 billion retrieval data points.
- **Notifications:** 1 million HTTPS notifications and 1000 email notifications per month.

Additional Services

- **Outbound Data Transfer:** 10 TB per month of outbound data.
- **Logging:** A fully managed, scalable log service for all logs in your tenancy.
- **Bastion:** Restricted, time-limited SSH access to target resources without public endpoints — free for both free and paid accounts.

Service Usage and Limits

You can check your tenancy's Always Free resource limits in the Console: **Governance & Administration** → **Tenancy Management** → **Limits, Quotas and Usage**.

Compute Shapes

A shape is a template that determines the number of OCPUs, amount of memory, and other resources allocated to an instance. Compute shapes are available with AMD, Intel, and Arm-based processors.

Note: When a new region becomes available, host capacity may take a few weeks to also become available. If your instance uses stateful security rules, each instance has a maximum number of concurrent connections that can be tracked, based on its shape.

Compute Shape Pricing — OCPUs and vCPUs

Oracle measures compute resources differently from most providers. The **Oracle CPU (OCPU)** represents physical CPU cores and is the unit of measurement for x86 CPUs (AMD and Intel) and Arm CPUs (OCI Ampere Compute). A **virtual CPU (vCPU)** — the industry-standard measurement — represents one execution thread of a physical CPU core.

Provisioning units for compute instances:

- 1 OCPU on Arm A1 (Compute) = 1 core on Arm A1 (Compute), or 1 vCPU
- 1 OCPU on Arm A2/A4 (Compute) = 2 cores on Arm A2/A4 (Compute), or 2 vCPUs
- 1 OCPU on x86 (AMD and Intel) = 2 vCPUs (most x86 architectures run 2 threads per physical core)

Flexible Shapes

A flexible shape lets you customize the number of OCPUs and amount of memory when launching or resizing your VM. Network bandwidth and number of VNICs scale proportionately with OCPU count.

The flexible shapes include: VM.Standard3.Flex (Intel), VM.Standard4.Ax.Flex (Intel), VM.Standard.E4.Flex / E5.Flex / E6.Flex / E6.Ax.Flex (AMD), VM.Standard.A1.Flex (Ampere Altra), VM.Standard.A2.Flex (AmpereOne), VM.Standard.A4.Flex / A4.Ax.Flex (AmpereOne M), VM.DenseIO.E4.Flex / E5.Flex / E6.Ax.Flex (AMD), and VM.Optimized3.Flex (Intel).

Note: Capacity might be limited for A1 shapes.

Maximum OCPU / Memory for Flexible Standard Shapes

Shape	Maximum OCPUs	Minimum Memory	Maximum Memory
VM.Standard3.Flex	32	1 GB or = OCPU count	64 GB/OCPU, up to 512 GB
VM.Standard4.Ax.Flex	39	1 GB or = OCPU count	64 GB/OCPU, up to 360 GB
VM.Standard.E4.Flex	64	1 GB or = OCPU count	64 GB/OCPU, up to 1024 GB
VM.Standard.E5.Flex	126	1 GB or = OCPU count	64 GB/OCPU, up to 1024 GB
VM.Standard.E6.Flex	126	1 GB or = OCPU count	64 GB/OCPU, up to 1454 GB
VM.Standard.E6.Ax.Flex94		1 GB or = OCPU count	64 GB/OCPU, up to 712 GB
VM.Standard.A1.Flex	76 (1 core/OCPU)	1 GB or = OCPU count	64 GB/OCPU, up to 472 GB
VM.Standard.A2.Flex	78 (2 cores/OCPU)	1 GB or = OCPU count	64 GB/OCPU, up to 946 GB

Shape	Maximum OCPUs	Minimum Memory	Maximum Memory
VM.Standard.A4.Flex	45 (2 cores/OCPU)	1 GB or = OCPU count	64 GB/OCPU, up to 700 GB
VM.Standard.A4.Ax.Flex	45 (2 cores/OCPU)	1 GB or = OCPU count	64 GB/OCPU, up to 720 GB
VM.Optimized3.Flex	18	1 GB or = OCPU count	64 GB/OCPU, up to 256 GB

Dense I/O Flexible Configurations

- **VM.DenseIO.E4.Flex:** 8 OCPU/128 GB, 16 OCPU/256 GB, or 32 OCPU/512 GB
- **VM.DenseIO.E5.Flex** and **VM.DenseIO.E6.Ax.Flex:** 8 OCPU/96 GB, 16 OCPU/192 GB, 24 OCPU/288 GB, 32 OCPU/384 GB, 40 OCPU/480 GB, or 48 OCPU/576 GB

These resources bill at per-second granularity with a one-minute minimum.

Virtual Machine (VM) Standard Shapes — Full Reference

Shape	OCPU	Memory (GB)	Local Disk	Max Network Bandwidth	Max VNICS (Linux)
VM.Standard3.Flex	32	1-512	Block storage only	1 Gbps/OCPU, max 32 Gbps	1 NIC/OCPU, max 24
VM.Standard4.A1-Flex	1-32	1-360	Block storage only	1 Gbps/OCPU, max 99 Gbps	1 NIC/OCPU, max 24
VM.Standard.E2.1.Micro (Always Free)	1		Block storage only	480 Mbps	1
VM.Standard.E4.Flex	1-64	1-1024	Block storage only	1 Gbps/OCPU, max 40 Gbps	1 NIC/OCPU, max 24
VM.Standard.E5.Flex	1-26	1-1049	Block storage only	1 Gbps/OCPU, max 40 Gbps	1 NIC/OCPU, max 24
VM.Standard.E6.Flex	1-26	1-1454	Block storage only	1 Gbps/OCPU, max 99 Gbps	1 NIC/OCPU, max 24
VM.Standard.E6.A1-Flex	1-24	1-712	Block storage only	1 Gbps/OCPU, max 99 Gbps	1 NIC/OCPU, max 24
VM.Standard.A1-Flex (Always Free pool)	1-472		Block storage only	1 Gbps/OCPU, max 40 Gbps	1 NIC/OCPU, max 24
VM.Standard.A2.Flex	1-78	1-946	Block storage only	1 Gbps/OCPU, max 78 Gbps	1 NIC/OCPU, max 24
VM.Standard.A4.Flex	1-45	1-700	Block storage only	1 Gbps/OCPU, max 100 Gbps	1 NIC/OCPU, max 24
VM.Standard.A4.A1-Flex	1-45	1-720	Block storage only	1 Gbps/OCPU, max 100 Gbps	1 NIC/OCPU, max 24

Processor details:

- VM.Standard.E2.1.Micro: AMD EPYC 7551 (2.0 GHz base / 3.0 GHz boost), AMD EPYC 7742 (2.25 GHz / 3.4 GHz), or AMD EPYC 7J13 (2.55 GHz / 3.5 GHz) — Oracle assigns one of these.
- VM.Standard.A1.Flex: Ampere Altra Q80-30, max 3.0 GHz, each OCPU = 1 hardware execution thread.
- VM.Standard.A2.Flex: Ampere AmpereOne A160-30, max 3.0 GHz, each OCPU = 2 hardware execution threads (2 cores).

GPU Shapes (VM)

Shape	OCPU	GPU Memory (GB)	CPU Memory (GB)	Max Network Bandwidth
VM.GPU2.1 (1x P100)	12	16	72	8 Gbps
VM.GPU3.1 (1x V100)	6	16	90	4 Gbps
VM.GPU3.2 (2x V100)	12	32	180	8 Gbps
VM.GPU3.4 (4x V100)	24	64	360	24.6 Gbps
VM.GPU.A10.1 (1x A10)	15	24	240	24 Gbps
VM.GPU.A10.2 (2x A10)	30	48	480	48 Gbps

Bare Metal Shapes — Standard Series

- **BM.Standard3.64:** Intel Xeon Platinum 8358 (2.6/3.4 GHz) — 64 OCPU, 1024 GB
- **BM.Standard4.Ax.120:** Intel Xeon 6987P-C (2.2/3.2 GHz) — 120 OCPU, 1152 GB
- **BM.Standard.E4.128:** AMD EPYC 7J13 (2.55/3.5 GHz) — 128 OCPU, 2048 GB
- **BM.Standard.E5.192:** AMD EPYC 9J14 (2.4/3.7 GHz) — 192 OCPU, 2304 GB
- **BM.Standard.E6.256:** AMD EPYC 9J45 (2.7/4.1 GHz) — 256 OCPU, 3072 GB, 2x 960GB NVMe
- **BM.Standard.A1.160:** Ampere Altra Q80-30, 3.0 GHz — 160 OCPU, 1024 GB (Windows not supported)
- **BM.Standard.A4.48:** Ampere AmpereOne M A06-36M, 3.6 GHz — 48 OCPU, 768 GB, 1x 3.84TB NVMe (Windows not supported)

Bare Metal GPU Shapes (selection)

Shape	OCPU	GPU Memory (GB)	CPU Memory (GB)
BM.GPU4.8 (8x A100)	64	320	2048
BM.GPU.A100-v2.8 (8x A100)	128	640	2048
BM.GPU.H100.8 (8x H100)	112	640	2048
BM.GPU.H200.8 (8x H200)	112	1128	3072
BM.GPU.B200.8 (8x B200)	128	1440	4096
BM.GPU.B300.8 (8x B300)	128	2100	4096
BM.GPU.GB200.4 (4x B200)	144	768	960
BM.GPU.MI300X.8	112	1536	2048
BM.GPU.MI355X.8	128	288	3072
BM.GPU.L40S.4	112	192	1024

Shape	OCPU	GPU Memory (GB)	CPU Memory (GB)
BM.GPU.RTXPRO.8	144	8x96	3072

C-States and Frequency Scaling

Modern CPUs transition to power-saving states (“c-states”) when idle or underutilized. C0 is the normal, fully active operating mode; higher c-states are deeper sleep modes that reduce energy use but increase wake-up latency, which can slow down demanding applications. The hypervisor on standard VM shapes manages this automatically: it prevents transitions to deeper sleep states even when the CPU is underutilized, and disables c-states entirely under sustained high utilization, keeping all cores active at base frequency (C0 state). The guest OS cannot manage c-states directly via kernel command-line options — it always reports the base frequency, even when the hypervisor is actually running the processor at its maximum advertised frequency.

Previous Generation Shapes

Previous generation shapes remain fully supported, but because the underlying hardware has reached the sustaining phase of its lifecycle, capacity in high-demand regions might be limited. To upgrade from a previous generation shape: for supported VM instances, change the instance’s shape directly; for bare metal instances or VMs that don’t support shape changes, terminate the instance **without deleting the boot volume**, then use that boot volume to launch a new instance on a current-generation shape.

Examples of previous-generation VM shapes (all with recommended current-generation replacements of VM.Standard.E4.Flex, E5.Flex, E6.Flex, A1.Flex, A2.Flex, or A4.Flex):

- **VM.Standard1** (end of orderability: Dec 31, 2020) — Intel Xeon E5-2699 v3, up to 16 OCPU / 112 GB
- **VM.Standard.B1** (end of orderability: Dec 31, 2020) — Intel Xeon E5-2699 v4, up to 16 OCPU / 192 GB
- **VM.Standard2** (end of orderability: Feb 28, 2022) — Intel Xeon Platinum 8167M, up to 24 OCPU / 320 GB
- **VM.Standard.E2** (end of orderability: Feb 8, 2021) — AMD EPYC 7551, up to 8 OCPU / 64 GB
- **VM.Standard.E3.Flex** (end of orderability: Mar 31, 2022) — AMD EPYC 7742, 1-64 OCPU flexible, 1-1024 GB

Dedicated Virtual Machine Host Shapes (selection)

Shape	Billed OCPU	Usable OCPU	Total Memory (GB)	Usable Memory (GB)
DVH.Standard3.64	64	60	1024	960
DVH.Standard.E4.128	128	124	2048	1912
DVH.Standard.E5.192	192	188	2304	2098

Fim do documento consolidado. Fonte original: Oracle Cloud Infrastructure Documentation (docs.oracle.com), páginas “Always Free Resources” e “Compute Shapes”.